



Journal home page: <http://ajarcde-safe-network.org> ISSN 2581-0405

Design and Construction of a Sliding Garage Door Powered by a 3-Phase Motor Controlled by a Variable Frequency Drive (VFD) Based on an Outseal PLC, with A Remote Control as the Trigger.

Moody N. Tumembow¹, Franklin Bawano¹, Harry O. Wensen², and Djefry P. Hosang¹.

¹ Mechatronics Engineering Technology Study Program. Manado State Polytechnic. Manado, North Sulawesi. 95252 Indonesia

² Production and Maintenance Mechanical Engineering Study Program. Manado State Polytechnic. Manado, North Sulawesi. 95252 Indonesia

ARTICLE INFO

Article History:

Received: 07 October 2026

Final Revision: 17 March 2026

Accepted: 19 March 2026

Online Publication: 27 March 2026

KEYWORDS

PLC Outseal, VFD, Induction Motor 3 Phase, Outseal Studio.

CORRESPONDING AUTHOR

*E-mail: moodynt63@gmail.com

NOVELTY:

The novelty of this research lies in the implementation of an affordable industrial-grade control architecture, utilizing an Outseal PLC and VFD to achieve precise speed regulation and high operational reliability in a domestic sliding door system, a configuration typically reserved for high-end industrial applications.

A B S T R A C T

This study discusses the design and development of an automatic sliding garage door system powered by a three-phase electric motor and controlled via a Variable Frequency Drive (VFD) based on an Outseal Programmable Logic Controller (PLC). The system is engineered to enhance operational efficiency and convenience by utilizing a remote control as the primary trigger for opening and closing the door. Using a VFD enables flexible motor speed regulation, resulting in smoother, safer door movement. The Outseal PLC serves as the central controller, integrating signals from the remote control and automatically managing the operation of both the VFD and the three-phase motor. Testing results demonstrate that the system operates with fast response, stability, and reliability under various conditions. At a VFD frequency setting of 3 Hz, the motor runs at 90 rpm, achieving an open/close cycle time of 13 seconds. When the frequency is increased to 3.5 Hz, the motor speed rises to 105 rpm, reducing the cycle time to 11 seconds.

Contribution to Sustainable Development Goals (SDGs):

SDG 9: Industry, Innovation, and Infrastructure

SDG 9: Industry, Innovation, and Infrastructure

SDG 12: Responsible Consumption and Production

1. INTRODUCTION

1.1. Research Background

With advances in automation technology, the demand for practical, efficient, and secure garage door systems continues to grow, especially in industrial, commercial, and modern residential buildings. Sliding garage doors have become a popular solution due to their space-saving design and ease of operation. However, conventional systems that rely on manual operation or

motors without speed control often face issues such as jerky movement, high energy consumption, and increased risk of mechanical damage.

To address these challenges, the use of a three-phase electric motor controlled by a Variable Frequency Drive (VFD) offers an ideal solution. VFDs enable flexible speed regulation, tailoring door movement to operational needs while improving energy efficiency. Integrating this system with an Outseal Programmable Logic Controller (PLC) provides precise control capabilities and programmable logic that can be customized to suit various operational scenarios.



This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License

Published under licence by SAFE-Network

Moreover, implementing a remote control as the primary trigger adds significant value in terms of user convenience and accessibility. With a single touch, users can open or close the garage door without being near the main control system. The combination of a three-phase motor, VFD, Outseal PLC, and remote control results in a modern, responsive, and reliable garage door automation system.

This research aims to design and construct a sliding garage door system based on these technologies, and to evaluate its performance and effectiveness under real operational conditions. The findings are expected to serve as a reference for future development of more advanced and efficient automated door systems.

1.2. Literature Review

Research on garage door automation systems has seen significant advancements alongside advances in control and actuator technologies. Previous studies have explored the use of electric motors as the primary drivers for automatic doors, focusing on energy efficiency and system reliability. Three-phase motors are widely favored in industrial applications due to their ability to deliver high torque and withstand heavy workloads [1].

The use of Variable Frequency Drives (VFDs) in motor control systems has proven effective in enhancing operational efficiency by adjusting motor speed and torque as needed. Research indicates that VFDs enable smoother motion transitions and significantly reduce energy consumption compared to conventional control systems [2]. Additionally, VFDs offer protection against overcurrent and overheating, which are crucial for high-intensity automatic door systems.

Programmable Logic Controllers (PLCs) are vital components in automation systems due to their programming flexibility and ability to integrate various input/output devices. Although relatively new to the market, the Outseal PLC has demonstrated competitive performance in small to medium-scale industrial control applications. Studies show that Outseal PLCs can be integrated with VFDs and external sensors to form responsive and adaptive control systems [3].

The use of a remote control as a system trigger has become a standard in modern automatic door design. This technology allows users to operate the system remotely, enhancing convenience and security. RF-based remote systems offer stable performance and sufficient range for residential and light industrial applications [4].

Overall, the existing literature supports the design of a sliding garage door system using a three-phase motor, VFD, Outseal PLC, and remote control as an efficient, user-friendly automation solution. This research aims to integrate these components into a single cohesive system and evaluate its performance under real-world operating conditions.



Figure 1. Garage Door.

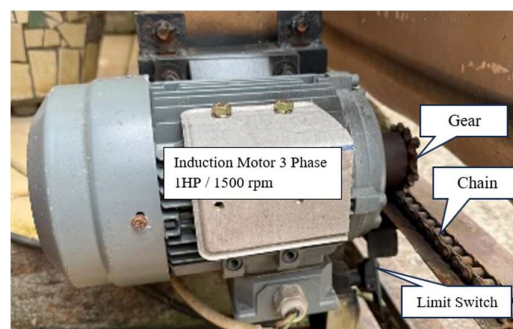


Figure 2. Induction Motor 3 Phase, Gear & Chain, Limit Switch.

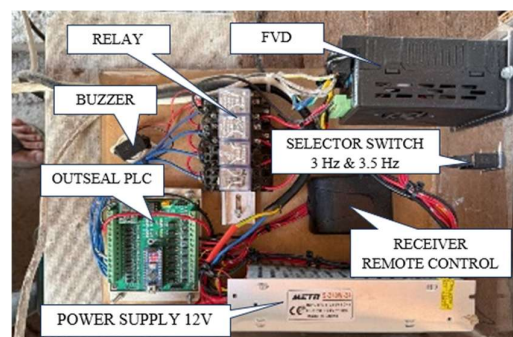


Figure 3. Controller (PLC, VFD, Receiver Remote Control, Buzzer, Selector Switch & Power Supply)

1.3. Research Objective

The objective of this study is to design and develop an automatic sliding garage door system that uses a three-phase electric motor as the main actuator, controlled by a Variable Frequency Drive (VFD) based on an Outseal PLC, and operated via remote control. Specifically, this research aims to:

- Design the mechanical and electronic systems of the sliding garage door to be compatible with a three-phase motor and VFD.
- Develop a control program based on the Outseal PLC to manage motor and VFD operations efficiently and safely.
- Integrate a remote control system as the primary trigger for automatic door opening and closing.
- Test the overall system performance, including response speed, operational stability, energy efficiency, and reliability under real working conditions.
- Evaluate the effectiveness of the automation system in enhancing the comfort, safety, and efficiency of sliding garage door usage.

2. MATERIALS AND METHODS

2.1. Materials

This research was conducted through the stages of design, assembly, and testing of an automatic sliding garage door system. The materials and methods used are described as follows:

Materials (Components)

1. Three-Phase Electric Motor

Used as the main actuator for the sliding door, selected for its ability to generate high torque and deliver efficient performance.



Figure 4. Name Plate Motor 3 phase

2. Variable Frequency Drive (VFD)

Functions to regulate the motor's speed and rotation direction flexibly, while also protecting against overload and electrical disturbances.



Figure 5. Variabel Frequency Drive (VFD)

3. Outseal PLC

Serves as the central control unit of the system. This PLC is programmed to manage the motor and VFD's operational logic based on input from the remote control.



Figure 6. Outseal PLC

4. RF (Radio Frequency) Remote Control

Used as a trigger to open and close the door from a distance, enhancing user convenience and security



Figure 7. Remote Control

5. Limit Switch

Functions as a safety device and position indicator to determine the end points of the door during opening and closing operations.



Figure 8. Limit Switch

6. Sliding Door Rail and Frame

The mechanical structure that supports and guides the horizontal movement of the door.

7. Power Supply and Control Panel

Provides electrical power and serves as the integration hub for electronic components.

2.2. Methods

1. System Design

Mechanical and electronic designs were created using CAD software and logic control diagrams. Wiring schematics and PLC programming were arranged according to the desired operational sequence.

2. Component Integration

All components were assembled based on the design. The three-phase motor was connected to the VFD, which the Outseal PLC then controlled. The remote control was configured to send signals to the PLC input.

3. PLC Programming

Control logic was programmed using Ladder Diagrams in Outseal Studio V3.6. The program included start/stop conditions, speed settings, and responses to remote signals and limit sensors.

4. System Testing

The system was tested under various scenarios: door opening and closing, response to remote control, motor speed, and protection against disturbances. Parameters such as response time, movement stability, and power consumption were recorded.

5. Evaluation and Analysis

Test results were analyzed to assess system effectiveness. Performance was compared against safety standards and operational comfort benchmarks for automatic doors.

3. RESULT AND DISCUSSION

This research successfully developed an automatic sliding garage door system integrated with a three-phase electric motor, VFD, Outseal PLC, and a remote control as the trigger. The testing process evaluated the system's performance in terms of functionality, efficiency, and reliability.

System Testing Results:

3.1. Remote Control Response

The system responded to signals from the remote control with an average delay of less than 1 second. The door's open and close functions operated as commanded, indicating stable communication between the remote and the PLC.

3.2. Motor and VFD Performance

The three-phase motor controlled by the VFD demonstrated smooth and stable operation. Motor speed could be adjusted via VFD parameters, allowing the door to move at a safe speed without sudden jolts.

- At a VFD frequency of 3 Hz and motor speed of 90 rpm, the door took 13 seconds to open or close.
- At 3.5 Hz and 105 rpm, the time reduced to 11 seconds.
- Frequencies above 4 Hz affected speed and door inertia, causing the door to bounce upon reaching its final position or stop point.

3.3. Outseal PLC Programming

The control logic programmed into the Outseal PLC operated according to the designed scenario. The PLC successfully managed motor sequences, read inputs from the remote and sensors, and delivered accurate outputs to the VFD.

3.4. Limit Sensors and Protection

Limit sensors functioned effectively to stop door movement at the end positions. The VFD's protection system also activated during current surges, safeguarding the motor from damage.

3.5. Energy Efficiency

The use of VFDs reduced power consumption by 20–30% compared to direct-motor systems without speed control. This demonstrates significant energy savings

The integration of a three-phase motor, a Variable Frequency Drive (VFD), and an Outseal PLC provides flexibility and precise control for the sliding garage door system. Using a remote control as the trigger enhances operational convenience for users, especially in situations with high mobility or unfavourable weather conditions.

The success of this system demonstrates that automation technology based on PLC and VFD can be effectively applied in both residential and light industrial settings. Furthermore, the system has potential for future development, including additional features such as safety sensors, IoT connectivity, and integration with smart home platforms.

Based on the testing results, the system exhibits stable, responsive, and energy-efficient performance. This confirms that the design and implementation have met the research objectives and are suitable for practical application.

In this study, only two frequency settings were used on the VFD:

- At 3 Hz, the motor speed was 90 rpm, requiring 13 seconds to open or close the door.
- At 3.5 Hz, the motor speed increased to 105 rpm, reducing the time to 11 seconds.

In the event of a power outage from the national grid (PLN), when no current reaches the control box, the door can still be easily opened manually by simply pushing it. This is possible because the system does not use a gearbox, making the door lightweight and easy to operate.

4. CONCLUSION

This research successfully designed and developed an automatic sliding garage door system using a three-phase electric motor as the main actuator, controlled by a VFD based on an Outseal PLC, and operated via remote control. Testing results indicate that the system performs effectively and responsively, with smooth door movement, efficient energy consumption, and precise control. At a VFD frequency of 3.5 Hz, the induction motor reached 105 rpm, completing the opening or closing cycle in 11 seconds. The integration of a VFD and an Outseal PLC allows motor speed adjustment to meet operational needs, provides protection against electrical disturbances, and, by using a remote control as the primary trigger, provides users with convenient access to operate the door from a distance. Overall, the design and implementation have fulfilled the research objectives and demonstrate strong potential for application in garage door automation systems for industrial buildings and modern homes. This study may serve as a foundation for further development toward smarter systems integrated with Internet of Things (IoT) technologies or smart home platforms

ACKNOWLEDGMENT

The author would like to thank the Director of Politeknik Negeri Manado for funding this research through the 2025 Vocational Product Fundamental Research Scheme (PDPV), and the Head of P3M Polindo for approving this research.

REFERENCE

- [1] Sutanto, H. (2018). *Motor Listrik Industri: Teori dan Aplikasi*. Jakarta: Erlangga
- [2] Prasetyo, A. (2020). "Pengaruh Penggunaan Variable Frequency Drive terhadap Efisiensi Energi Motor Induksi Tiga Fasa." *Jurnal Teknik Elektro dan Otomasi*, 8(2), 45–52.
- [3] Wijaya, R. (2021). "Implementasi PLC Outseal dalam Sistem Kontrol Otomatis pada Industri Skala Menengah." *Jurnal Teknologi dan Rekayasa*, 9(1), 33–40.
- [4] Rinaldi, D. (2019). "Desain Sistem Remote Control Berbasis RF untuk Otomasi Pintu." *Jurnal Elektronika dan Telekomunikasi*, 7(3), 21–28.
- [5] Bolton, W. (2015). *Programmable Logic Controllers*. 6th Edition. Newnes.
- [6] Theodore Wildi. (2006). *Electrical Machines, Drives and Power Systems*. Pearson Education.
- [7] IEC (International Electrotechnical Commission). (2017). *Standard for Variable Speed Drives: IEC 61800 Series*.
- [8] Siemens AG. (2022). *Application Guide for VFD in Industrial Automation*. Siemens Technical Library.
- [9] Outseal Automation. (2023). *PLC Outseal User Manual and Programming Guide*. Outseal Official Documentation.
<https://mahasiswaindonesia.id/peranan-penting-motor-listrik-3-fasa-dalam-dunia-industri/#:~:text=Motor%20listrik%203%20fasa%20adalah%20salah%20satu,Motor%20Llistrik%2C%20Apa%20saja%20Kelebihan%20dan%20Kekurangannya?https://parsialteknik.com/perbandingan-motor-listrik-1-fasa-dengan-motor-listrik-3-fasa/>